Answers and marking guidance

Award one mark per correct, developed point unless stated. Other valid answers should be credited.

2 State the Boolean operation that keeps only the volume where two solids overlap.

- Intersect

3 Quick check

- a) Intersect
- b) Centre axis and angle
- c) Loft
- d) Revolve

4 Key terms

- 1 – A
- 2 – D
- 3 – E
- 4 – B
- 5 – F
- 6 – C

5 Describe how the model should be changed.

- Select the profile (sketch) on work plane 2 in the modelling tree
- Edit it: replace the ellipse with a hexagon of 24 across flats; the model updates

No marks for: Remodelling from scratch scores 0

6 Describe two different ways the curved top could be created in 3D CAD.

- Extrude method: sketch the profile on a face or work plane, extrude, then fillet the top edge
- Revolve method: sketch the profile and revolve 90° about an axis
- Sizes or angle given

7 Describe how the spring would be modelled.

- Ø3 profile positioned 25 from the centre axis
- Use the helix command
- Any two of: 11 revolutions, pitch 20, height 220